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$$I = \int_0^1 \frac{x^3}{1+x^4} dx$$

$$\text{Stel } t = 1 + x^4$$

$$\Rightarrow dt = 4x^3$$

$$\Rightarrow \frac{1}{4} dt = x^3$$

$$\begin{aligned} I &= \frac{1}{4} \int_1^2 \frac{1}{t} dt \\ &= \frac{1}{4} [\ln |t|]_1^2 = \frac{1}{4} \ln 2 \end{aligned}$$

References